Foodbank Fridge

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Executive Summary

Problem

Restaurants, caterers, and campus dining operations throw away large amounts of edible prepared food every day, mostly because they have no quick and safe way to get it to people who could eat it. At the same time, many households in our own city skip meals or rely on food pantries that keep limited hours and often lack fresh or prepared items. Community refrigerators have appeared in some neighborhoods as a response, but most are informal. They are stocked unpredictably, nobody monitors whether food has stayed cold, and neighbors cannot tell in advance whether anything is inside. Donors worry about liability and about whether their food will reach anyone at all. The result is that surplus food and hungry neighbors exist within a few blocks of each other and still do not connect.

Solution

Foodbank Fridge combines a physical refrigerator, a software layer, and a routing service. The refrigerator is a commercial-grade unit paired with a small solar panel and battery that offset its power draw and keep it running through short outages, which lowers the cost of hosting it. Sensors inside each unit record temperature and door activity and send readings to our dashboard every few minutes. If the unit leaves the safe range for too long, the system alerts a local volunteer steward and marks the contents for removal.

The app serves three groups. Donors, meaning restaurants and caterers, post available food with a short description and a pickup or drop-off window. Our routing service matches the donation to the nearest fridge with space and notifies nearby registered neighbors. Neighbors browse what is currently available, reserve items, and pick them up without needing to prove need or fill out paperwork. Stewards, typically staff or volunteers at the host site, confirm deliveries and check the unit.

Market

We see three groups of customers. The first is food donors. The United States has several hundred thousand restaurants, and a meaningful share of them have regular surplus. Our initial focus is independent restaurants and caterers within a few miles of campus. The second is host sites such as community centers, houses of worship, and housing complexes that want a food resource without running a full pantry. The third is institutions, such as city agencies, health systems, and universities, that fund food access and sustainability programs and need reporting on results. We have not completed a formal market sizing. Using rough national figures for restaurant count and food waste, we believe the long-term opportunity is large, but our near-term market is the set of restaurants and host sites in our metropolitan area, which we estimate at a few hundred potential donors.

Business Model

We plan to earn revenue from several sources. Donors would pay a monthly subscription for the app, routing, and a donation record they can use for tax documentation and sustainability reporting. Host sites or sponsors would pay a placement fee or a lease for the fridge unit. Institutional partners would pay for data reports on food rescued, locations served, and temperature compliance. We also expect to apply for grants during the early period. Our working assumption is a donor subscription in the range of twenty to forty dollars per month and a sponsor fee of a few hundred dollars per unit per year, though we have not yet tested what customers will actually pay. At present the service is free to everyone, so we have no revenue. A key open question is whether restaurants see enough value to pay, since many already donate informally at no cost.

Competition

The space is crowded. Food rescue organizations and apps already connect restaurants and grocers with charities, and some work at large scale. Traditional food banks and pantries have established relationships, trucks, and funding. Community fridge networks run by volunteers offer a similar physical presence at near-zero cost. Surplus-food marketplaces sell discounted food to consumers rather than giving it away. We think our difference is the combination of monitored cold storage, neighborhood-level routing, and a safety record in a single product, which most of these alternatives do not offer together. We recognize that a larger organization could add similar features, and that volunteer-run fridges may be good enough for many neighborhoods. Our advantage today is speed and a focus on temperature safety, not any protected technology.

Team

Marisol Venkataraman, chief executive, studies public health and leads partnerships with restaurants and host sites. She volunteered for two years at a campus food pantry. Dexter Oyelaran, chief technology officer, studies computer science and built the app, the sensor data pipeline, and the routing logic. Priya Halloran-Wu, chief operating officer, studies mechanical engineering and designed the solar and battery configuration and the fridge enclosure. We have a faculty advisor in the school of public health who has offered feedback on food safety practice, and we are looking for additional mentors with experience in food service regulation and nonprofit partnerships. All three of us are full-time students and are working on the company alongside coursework.

Milestones and Financials

We have defined the core concept, formed a Delaware C corporation, and launched a basic website. We built a working prototype of the fridge with solar assist and sensors, and the first unit has been operating at a partner community center for about six weeks. During that time, a handful of restaurants and several dozen neighbors have signed up and used the app. We have logged the temperature readings and recorded a small number of alerts, all of which were resolved without food loss. We do not yet have a rigorous count of meals rescued, and our early user numbers are small and come largely from people we recruited personally.

So far we have spent roughly six thousand dollars, funded by personal savings and a small student innovation grant, mostly on the fridge, panel, battery, and sensors. Our estimated cost for each additional unit is between three and four thousand dollars including installation. We expect to need around twenty-five thousand dollars over the next year to build several more units, improve the app, and pay for insurance and basic legal work. Revenue projections beyond that point depend on assumptions we still need to validate, so we are not presenting detailed forecasts in this summary.

What We Are Asking For

We are applying for prize funding to build a second and third unit and extend our pilot to more neighborhoods. Beyond money, we would value introductions to restaurant owners and health department contacts, advice on liability and food donation law, and feedback from judges on pricing. Over the coming months, our goal is to show that donors will pay for the service, that neighbors return to use it, and that we can operate several units safely without a large volunteer effort. We believe the competition would help us move from an interesting prototype toward a business with evidence behind it.